

FINAL PROJECT REPORT DOCSHARE

Web-Based Academic Resource Sharing Platform

Project Theme:	Academic Content Management & Student Collaboration
Group Number:	Group No. 22
Group Members:	• Member 1: U.H.N.D.P.Senarthna (Index No: 2785) • Member 2: R.M.V.N.Rathnyaka (Index No: 2774)
Academic Term:	ICT 1209 - Web Technologies Module
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2. PROJECT OVERVIEW

2.1 Purpose & Background

DOCSHARE serves as a centralized digital repository designed to eliminate the fragmentation of academic study resources among undergraduates. Traditionally, lecture slides, past examination papers, lab manuals, and revision guides are scattered across social media group chats, temporary messaging threads, and personal cloud storage folders. This fragmentation leads to lost files, difficulty in discovering past materials, and unequal access to study resources.

DOCSHARE resolves these challenges by providing an organized, centralized platform where students can seamlessly search, filter, view, upload, download, and bookmark peer-contributed study materials.

Core Mission: To establish a structured, open-access academic library that fosters collaborative learning, streamlines resource discovery across BICT subjects, and provides productivity tools like personal study reminders.

2.2 Target Audience

BICT / ICT Undergraduates

Students seeking instant access to organized lecture slides, previous exam papers, lab tutorials, and subject-specific notes to support their studies.

Student Contributors & Reps

Academic batch representatives and high-achieving students who wish to share structured notes, solutions, and helpful guides with their peers.

2.3 Key Features & Functional Modules

- **User Authentication & Session Management:** Secure registration, encrypted login handling, and role-based session control (``require_login()``) protecting private routes.
- **Resource Directory & Advanced Search:** Filter documents by keyword title, subject code, file extension (PDF, DOCX, PPTX), or author.
- **File Upload & Categorization:** Upload engine with validation and metadata tagging (Title, Subject, Extension, Author, Description).
- **Bookmarks System:** One-click bookmarking tool allowing logged-in students to save documents to a personalized dashboard for instant reference.
- **Personalized Reminders Tracker:** Integrated event tracker where students can set study schedules, revision deadlines, or assignment reminders linked to documents.
- **Report & Feedback Handling:** Integrated issue reporting mechanism sending user feedback and broken link reports directly into the database (``reports`` table).

3. SYSTEM DESIGN

3.1 Relational Database Schema (MySQL)

The platform relies on a normalized relational database (`docshare\_db`) built in MySQL. Foreign key constraints ensure referential integrity when user accounts or uploaded documents are updated or removed.

```
CREATE DATABASE IF NOT EXISTS docshare_db;
USE docshare_db;

-- 1. Users Table
CREATE TABLE users (
  id INT AUTO_INCREMENT PRIMARY KEY,
  username VARCHAR(50) NOT NULL UNIQUE,
  email VARCHAR(100) NOT NULL UNIQUE,
  password VARCHAR(255) NOT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 2. Resources Table
CREATE TABLE resources (
  id INT AUTO_INCREMENT PRIMARY KEY,
  user_id INT NOT NULL,
  title VARCHAR(255) NOT NULL,
  subject VARCHAR(100) NOT NULL,
  file_type VARCHAR(20) NOT NULL,
  author VARCHAR(100) NOT NULL,
  description TEXT,
  file_path VARCHAR(255) NOT NULL,
  download_count INT DEFAULT 0,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
);

-- 3. Bookmarks Table
CREATE TABLE bookmarks (
  id INT AUTO_INCREMENT PRIMARY KEY,
  user_id INT NOT NULL,
  resource_id INT NOT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE,
  FOREIGN KEY (resource_id) REFERENCES resources(id) ON DELETE CASCADE
);

-- 4. Reminders Table
CREATE TABLE reminders (
  id INT AUTO_INCREMENT PRIMARY KEY,
  user_id INT NOT NULL,
  resource_id INT NULL,
  title VARCHAR(255) NOT NULL,
  remind_at DATETIME NOT NULL,
  note TEXT,
  status ENUM('pending', 'completed') DEFAULT 'pending',
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE,
  FOREIGN KEY (resource_id) REFERENCES resources(id) ON DELETE SET NULL
);

-- 5. Reports Table
CREATE TABLE reports (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100) NOT NULL,
  email VARCHAR(100) NOT NULL,
  subject VARCHAR(150),
  message TEXT NOT NULL
```

3.2 Application Architecture & Directory Structure

The project follows a modular Model-View-Controller (MVC) inspired separation of concerns for vanilla PHP web applications. Shared components (navigation headers, footers, database scripts) are isolated in dedicated subdirectories.

```
docshare/
├── config/
│   └── db.php           # Database connection configuration & PDO/MySQLi handles
├── uploads/            # Secured directory storing uploaded documents (PDF, DOCX, PPTX)
├── css/
│   ├── style.css       # Global typography, color variables, grid layouts
│   ├── contact.css     # Form controls and issue submission styles
│   └── nav.css          # Responsive navigation header & menu dropdowns
├── includes/
│   ├── nav.php         # Reusable global header component with dynamic active flags
│   └── footer.php      # Reusable global footer component
├── functions.php       # Core helpers: sanitize(), require_login(), auth checks
├── index.php           # Landing page with hero banner & quick search bar
├── browse.php          # Document repository search engine & filter interface
├── upload.php          # File upload drag-and-drop form
├── features.php        # Feature navigation hub (Bookmarks, Search, Reminders)
├── bookmarks.php       # User's saved bookmarks management dashboard
├── reminders.php       # User's active study reminders & calendar deadlines page
├── about.php           # Project information & background overview page
├── contact.php         # User support & issue reporting form page
└── logout.php          # User session cleanup and redirection handler
```

Modular Implementation Note: By referencing `includes/nav.php` and `functions.php` across all page scripts, structural site modifications or security updates can be propagated globally across the platform.

4. IMPLEMENTATION & MODULE WALKTHROUGH

4.1 Home Page (`index.php`)

The landing page serves as the entry point, featuring a hero section and search interface allowing students to immediately search for subject materials across BICT courses.

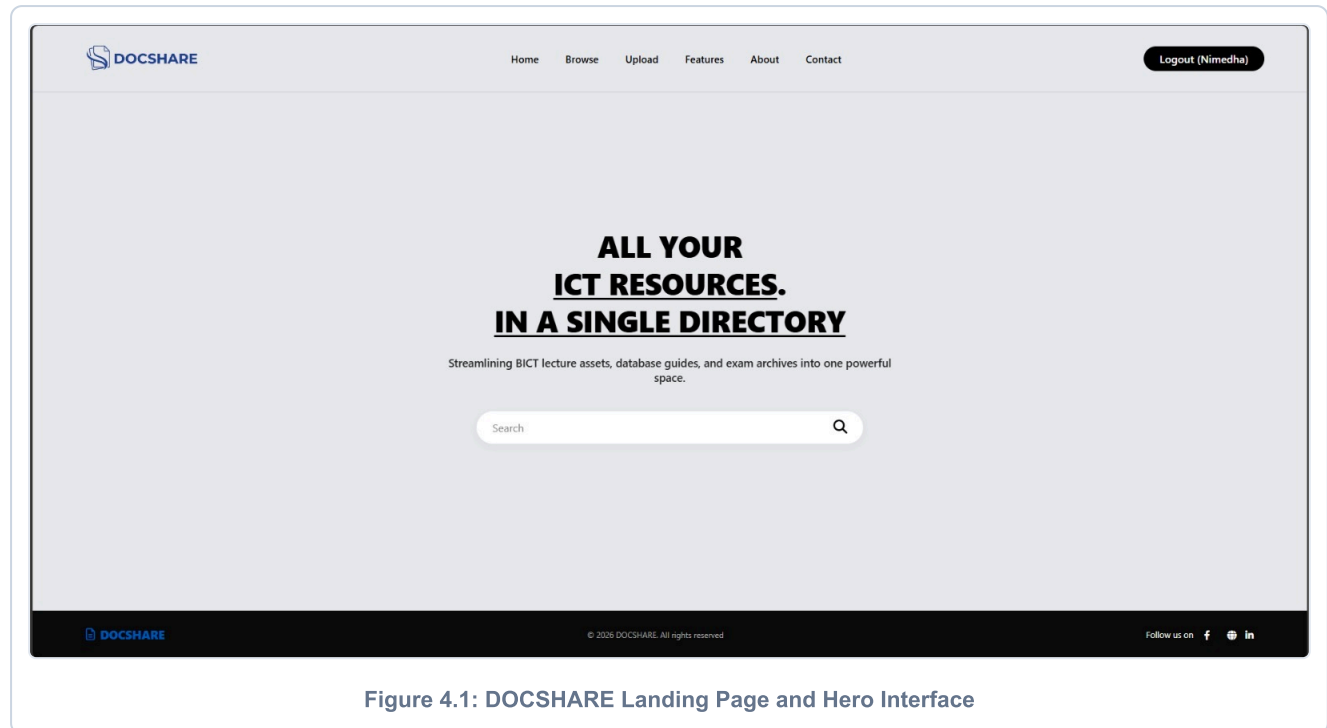


Figure 4.1: DOCSHARE Landing Page and Hero Interface

4.2 Browse Academic Resources (`browse.php`)

The repository interface features multifaceted filtering. Users can narrow down files by subject, format, or search query.

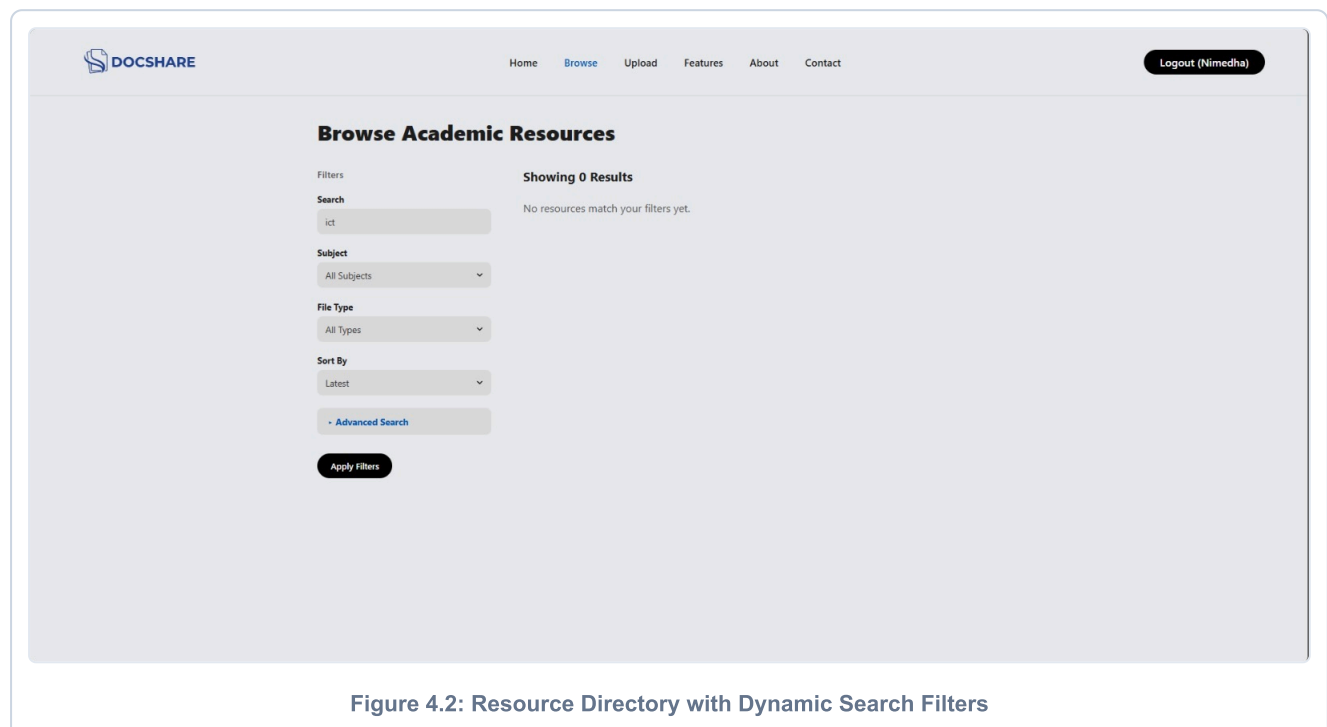


Figure 4.2: Resource Directory with Dynamic Search Filters

4.3 Upload Resource (`upload.php`)

The upload module allows authorized users to post study resources. File types are restricted, and uploaded files are saved into `/uploads/` with sanitized references stored in the `resources` table.

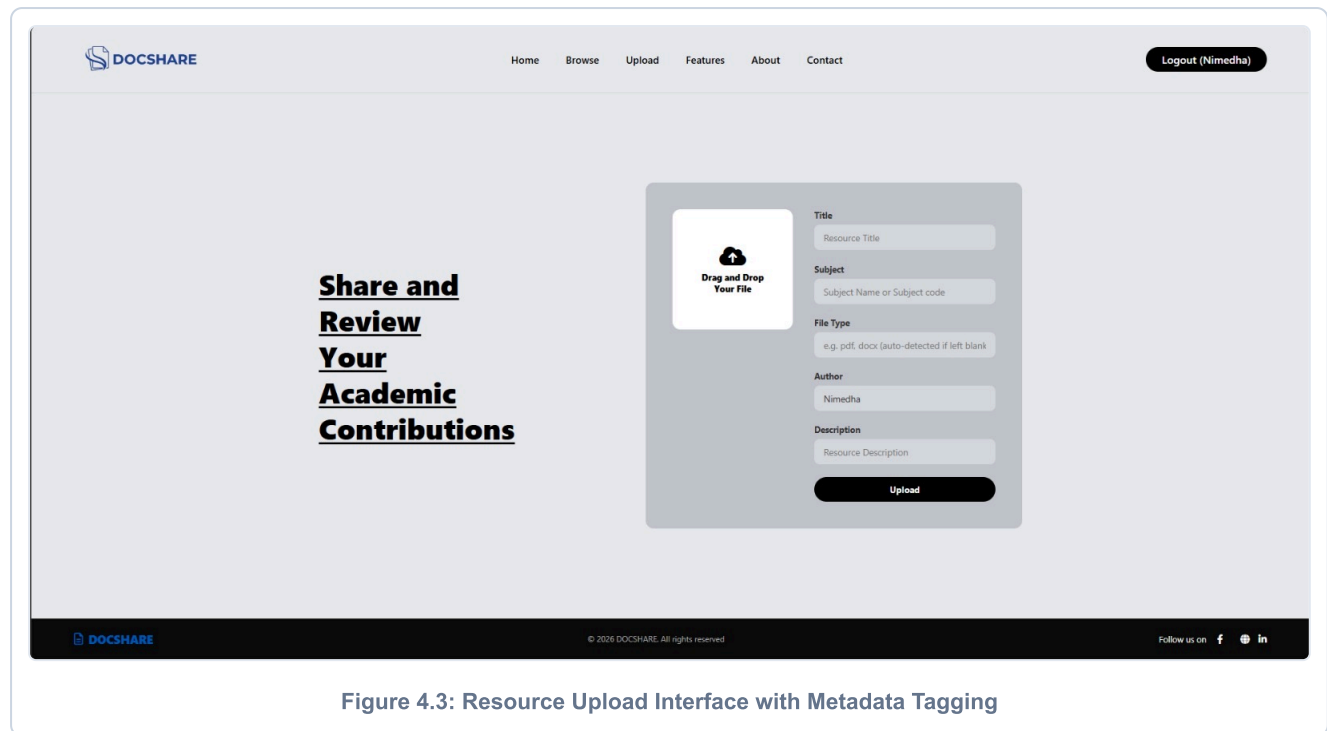


Figure 4.3: Resource Upload Interface with Metadata Tagging

4.4 Features Overview (`features.php`)

A dashboard guiding users through platform tools including Bookmarks, File Discovery, and Reminders.

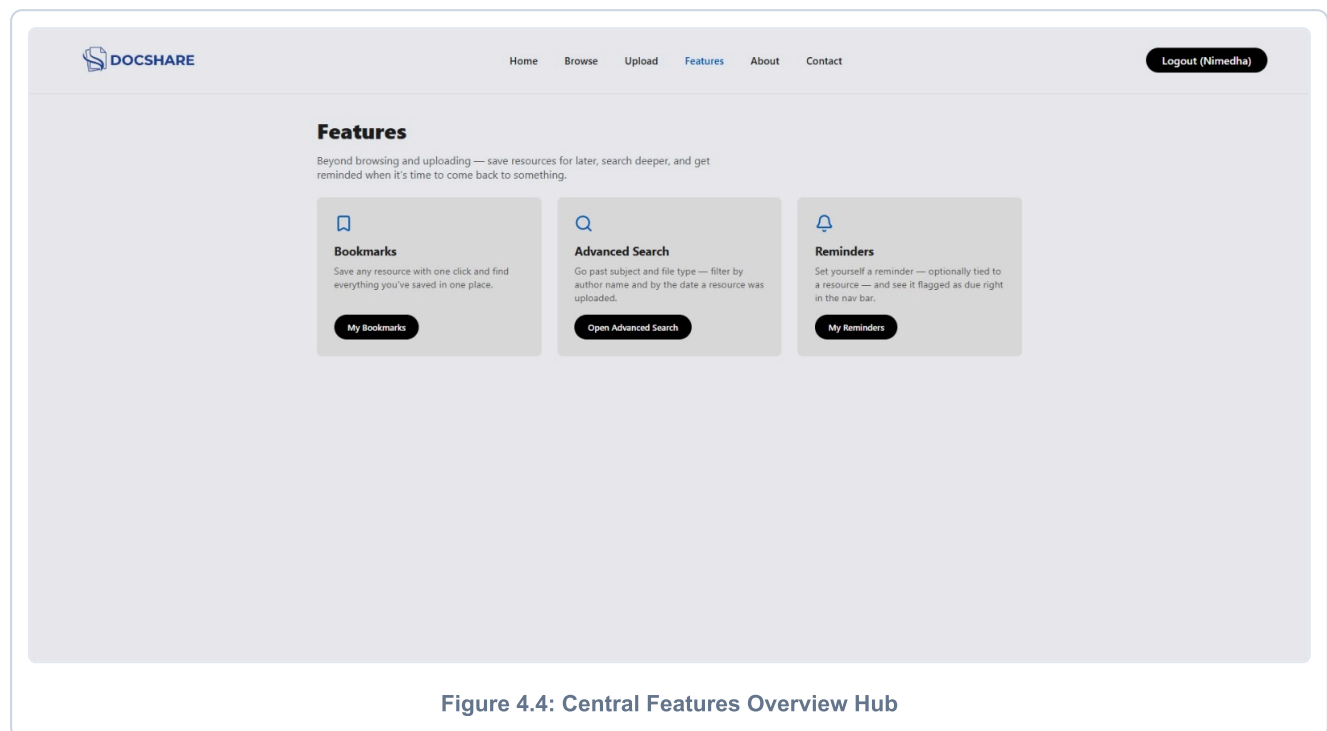


Figure 4.4: Central Features Overview Hub

4.5 My Bookmarks Dashboard (`bookmarks.php`)

Displays saved resources for quick access. Bookmarks can be managed or removed directly from this view.

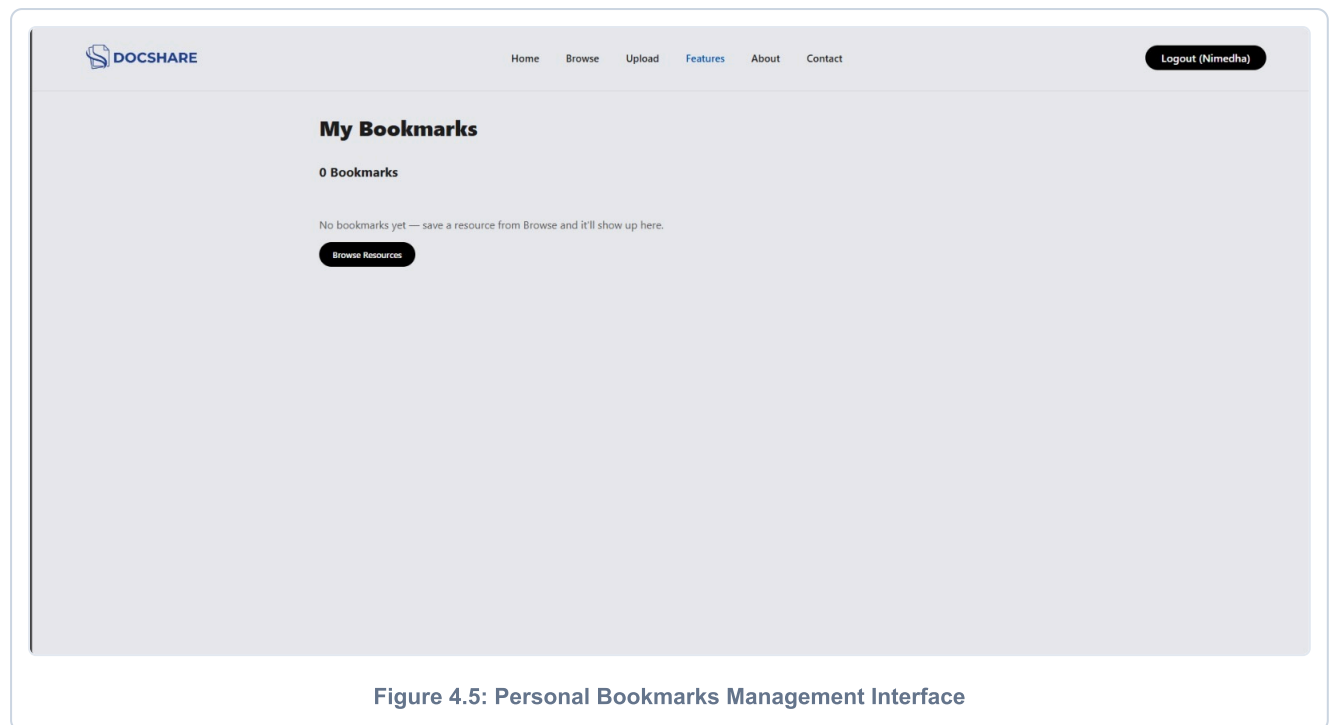


Figure 4.5: Personal Bookmarks Management Interface

4.6 Personal Reminders Tracker (`reminders.php`)

Enables students to set deadline alerts and study reminders tied to academic resources.

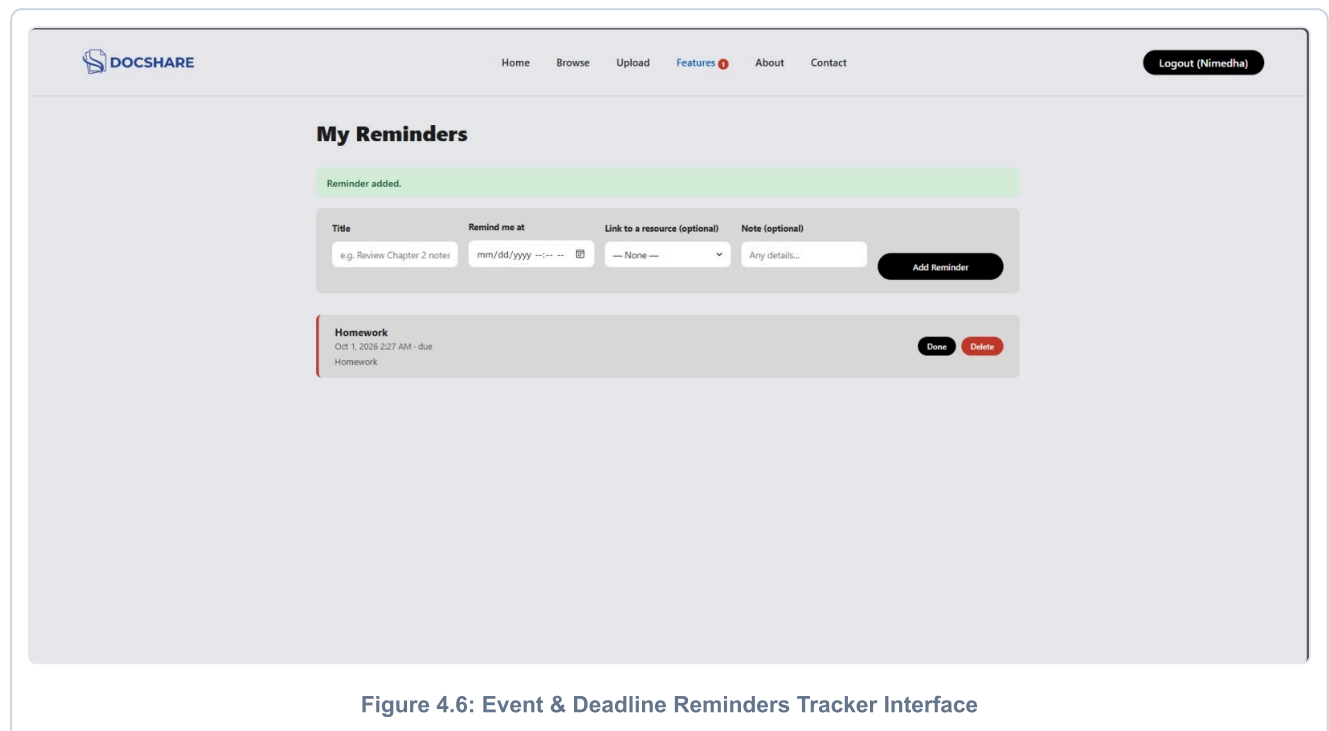


Figure 4.6: Event & Deadline Reminders Tracker Interface

4.7 About DOCSHARE (`about.php`)

Presents project objectives, underlying technologies, and team background for Group No. 22.

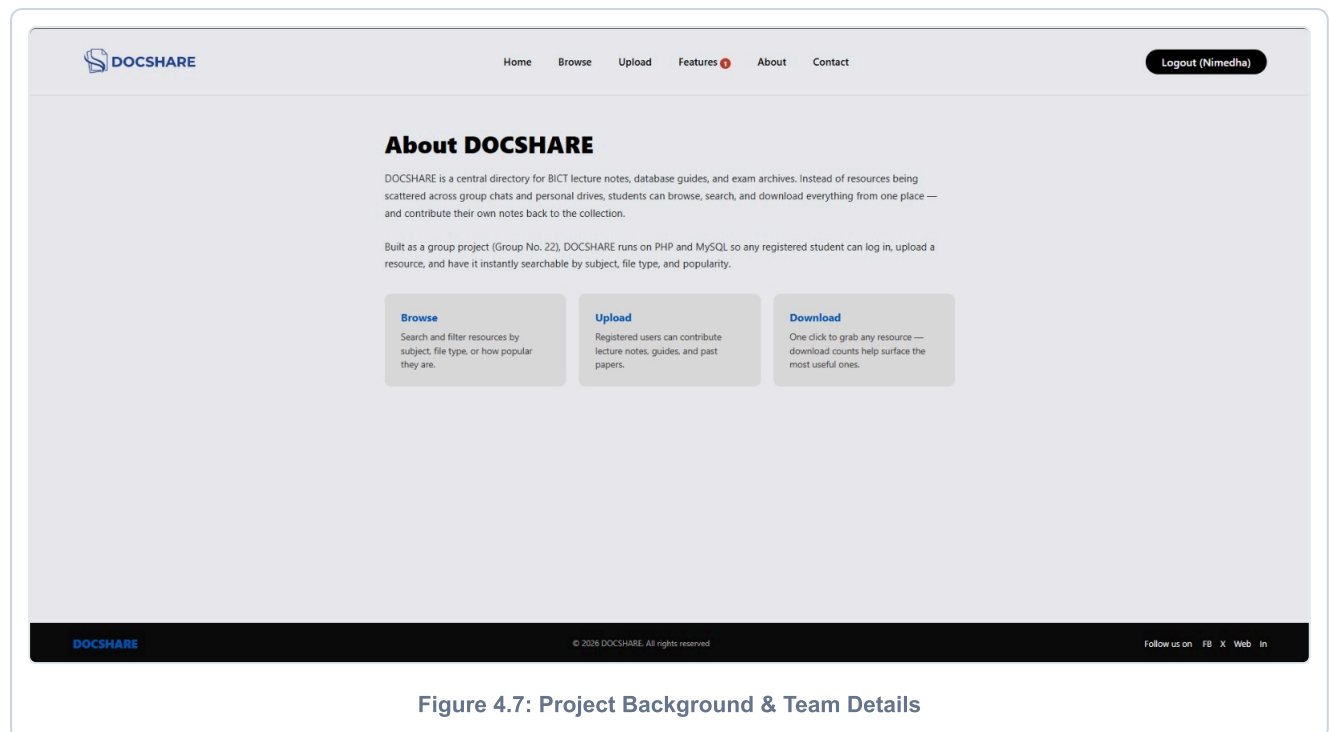


Figure 4.7: Project Background & Team Details

4.8 Contact & Issue Reporting (`contact.php`)

Allows users to report issues or submit feedback. Submissions are processed and stored in the database.

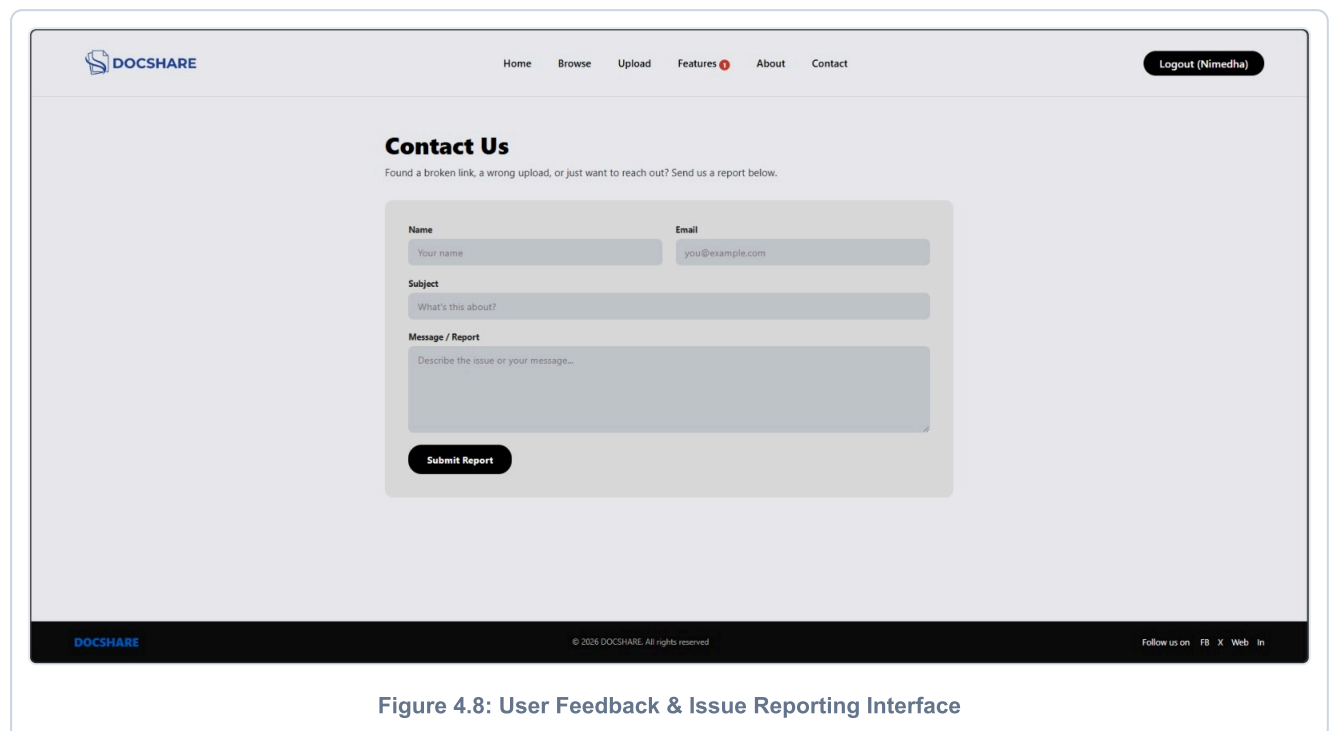


Figure 4.8: User Feedback & Issue Reporting Interface

5. TECHNICAL CHALLENGES AND SOLUTIONS

During development and deployment, several technical issues were addressed to ensure security, data integrity, and usability.

Technical Challenge	Risk / Impact	Technical Solution Applied
SQL Injection Vulnerabilities	Malicious input via search boxes or forms (e.g., `contact.php`) risking database access.	Wrapped user inputs in a centralized `sanitize()` helper function using `mysqli_real_escape_string()` and input trimming across POST handlers.
Orphaned File Storage	Uploaded files saved to the server without complete metadata entries in the database.	Implemented strict server validation (`\$_FILES` error checking) prior to calling `move_uploaded_file()`.
Unauthenticated Access	Direct URL access to protected pages like `reminders.php` or `bookmarks.php`.	Created a mandatory `require_login()` session verification helper called at the start of all protected scripts.
Dynamic Nav Highlighting	Navigation header failing to reflect active page states.	Passed dynamic `\$active` flag variables prior to rendering `nav.php` (e.g., `\$active = 'contact';`).
Cross-Site Scripting (XSS)	Unsanitized user comments or descriptions rendering malicious HTML/JS.	Escaped all database outputs using `htmlspecialchars()` before displaying content in the UI.

Security Implementation Summary: The platform combines input sanitization, dynamic query escaping, session checking, and output encoding to protect user data.

6. INDIVIDUAL TEAM CONTRIBUTIONS

Tasks were divided equally among group members based on system modules and development layers.

Student Name	Primary Responsibilities & Modules	Contribution
U.H.N.D.P.Senarthna	Frontend layout design, master CSS framework, navigation (`nav.php`), Features Hub (`features.php`), Reminders tracker (`reminders.php`), Resource search engine (`browse.php`), and filter queries.	50%
R.M.V.N.Rathnyaka	Database schema design (`docshare_db`), User Authentication engine (Login/Register/Logout), file upload pipeline (`upload.php`), Bookmarks system (`bookmarks.php`), and Contact submission handling (`contact.php`).	50%

7. SETUP INSTRUCTIONS (XAMPP DEPLOYMENT)

7.1 Prerequisites

- XAMPP Server (Apache Web Server + MySQL/MariaDB running PHP 8.x)
- Modern web browser (Google Chrome, Mozilla Firefox, Microsoft Edge)

7.2 Installation Steps

1. **Extract Project Files:** Place the `docshare` directory into the XAMPP web root:
`C:\xampp\htdocs\docshare`.
2. **Start XAMPP Services:** Open the XAMPP Control Panel and start Apache and MySQL.
3. **Import Database:**
 - Navigate to `http://localhost/phpmyadmin/`.
 - Create a database named `docshare\_db`.
 - Select `docshare\_db`, click **Import**, select `docshare\_db.sql`, and click **Go**.
4. **Configuration Check:** Verify that `config/db.php` has default local credentials (`root` user with no password).
5. **Launch Application:** Open browser and navigate to `http://localhost/docshare/index.php`.

8. REFERENCES & ACADEMIC SOURCES

- MySQL 8.0 Reference Manual. (2026). *Foreign Keys and Relational Integrity Constraints*. Oracle Corporation. Retrieved from <https://dev.mysql.com/doc/refman/8.0/en/>
- PHP Manual. (2026). *Session Handling, Data Sanitization, and File Upload Handling Functions*. The PHP Documentation Group. Retrieved from <https://www.php.net/manual/en/>
- W3Schools Online Web Tutorials. (2026). *PHP MySQL Database Integration & Secure Form Processing*. Refsnes Data. Retrieved from <https://www.w3schools.com/php/>
- MDN Web Docs. (2026). *HTML5 Dynamic Forms, CSS Flexible Box Layout, and Accessibility Standards*. Mozilla Foundation. Retrieved from <https://developer.mozilla.org/>